

# SAILAKSHMI B T

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## PROFESSIONAL SUMMARY

Robotics and Automation undergraduate (CGPA 8.99) at Madras Institute of Technology with hands-on experience in embedded systems, robotics, and AI-driven applications. Skilled in Arduino-based system design, sensor interfacing, and rapid prototyping. Interests span Computer Vision, industrial automation, embedded intelligence, and intelligent systems.

## EDUCATION

**B.E. – Robotics and Automation** – Madras Institute of Technology 2024 – 2028

- CGPA: 8.99 / 10.0 (current)
- Higher Secondary Certificate (2024): 94.16% | Secondary School Certificate (2022): 95.2%
- JEE (Main) 2024: 81.82 percentile (CRL: 269,607 | OBC-NCL: 94,238)

## INTERNSHIPS & TRAINING

**Computer Vision & AI Intern** – PixelBotix June 2026

- Completed hands-on training in image processing, object detection, and AI-based visual recognition. Developed and deployed small-scale projects applying computer vision and AI techniques.

**Industrial Robotics & Automation Training** – Birla Carbon & KUKA Feb 2026

- Trained on industrial robotic arm programming, automation workflows, and safety practices in a live manufacturing environment. Exposed to Industry 4.0 concepts and KUKA robot operation.

## PROJECTS

**Student Life Concierge AI Agent** – Google ADK, Gemini API, FastAPI, Python, Pytest, Uvicorn | Kaggle & Google 5-Day AI Agents Intensive

- Built a multi-agent FastAPI backend that parsed academic emails and automated student scheduling tasks using Google's Agent Development Kit and Gemini API.
- Added a prompt injection / anti-jailbreak layer to prevent the agent from acting outside its defined scope.
- Wrote a Pytest suite with mock services to evaluate agent behavior locally without live API calls.

**Masters Application Tracker & Resume Analyzer** – React, Vite, PDF.js, LocalStorage API, CSS3

- Developed a privacy-first, fully client-side platform for managing Master's application workflows.
- Built a PDF.js-based resume analysis engine for keyword extraction and profile evaluation.
- Designed local data management architecture and is further expanding the platform with a research planner for universities and academic programs.

**Orion – Local AI Assistant** – Python, Llama 3.2, Ollama, FastAPI, WebSockets, Web Speech API

- Developed an offline voice assistant powered by Llama 3.2 running locally through Ollama.
- Built an asynchronous FastAPI backend with WebSocket-based streaming.
- Currently expanding tool-calling capabilities for local system diagnostics.

**Single-Axis Gimbal Stabilization System** – Arduino Nano, MPU-6050, SG90 Servo

- Built a single-axis stabilization gimbal using an MPU-6050 IMU feeding a closed-loop PID controller that drives an SG90 servo motor in real time.
- Implemented inertial sensing and feedback loop entirely in embedded C on Arduino Nano; achieved stable angular correction under manual disturbance.

**Grip Stability & Stress Analyzer** – Arduino Uno, HX711, Load Cell

- Built a load-cell based measurement system using an HX711 ADC and Arduino Uno to capture grip force and tremor patterns over time for stress-level estimation.

**Adaptive Self-Mutating Authentication System** – 8051 (AT89S52), Embedded C, Hardware Timers

- Designed a digital lock on an 8051 microcontroller that regenerates a non-deterministic password after each login using hardware timers, making it resistant to replay attacks and shoulder-surfing.

## Mechanical Rotary Device – SolidWorks | Design Patent Pending

- Designed a dual-mode rotary mechanism combining a Geneva wheel and dog clutch in SolidWorks; filed for a design patent.

## HACKATHONS & COMPETITIONS

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### Smart India Hackathon 2025 – Delhi, Finalist

2025

- Worked within a cross-functional team to develop an AI/ML-based system for identifying cryptographic primitives and protocols in multi-architecture firmware binaries, focusing on the reverse engineering workflow and data analysis.

### H2TS Hackathon 2025 – 2nd Place

2025

- Collaborated with my team to build an AI-driven SOS platform for emergency response coordination; placed 2nd out of the competing teams.

### VISA 24-Hour Hackathon 2026 – Participant

2026

- Competed in a 24-hour sprint focused on fintech innovation.

## TECHNICAL SKILLS

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**Embedded Systems:** Arduino Uno R3, Arduino Nano, 8051 (AT89S52), Sensor Interfacing, Hardware Timers, I2C/SPI communication

**Control Systems:** PID Control, Closed-Loop Feedback, Motion Control, Inertial Sensing, Stabilization Mechanisms

**Programming:** Python (Intermediate), Embedded C (Fundamentals), MATLAB/Simulink (Intermediate)

**AI & Agentic Systems:** Google Agent Development Kit (ADK), Gemini API, Ollama, Llama 3.2, FastAPI, Pytest, Prompt Engineering

**Computer Vision:** Image Processing, Object Detection, AI-based Visual Recognition (OpenCV-level coursework)

**Design & EDA:** SolidWorks, KiCAD, Canva

**Cybersecurity / RE:** Binwalk, Ghidra, Firmware Binary Analysis, Ubuntu (working knowledge of tools and workflow)

## CERTIFICATIONS

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- AI Capabilities and Limitations – Anthropic
- Foundations of Prompt Engineering – AWS Training & Certifications
- Automation in Manufacturing – NPTEL (Elite, 67%)
- Python for Absolute Beginners – Udemy
- Basics of HTML – Udemy

## LEADERSHIP & ACTIVITIES

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- Placement Representative (2026 – Present): Liaison between students and placement coordinators for internship and placement communications.
- Joint Treasurer, MIT Quill (2025 – 26): Managed treasury for the college literary club.
- Office Bearer and Content writer, SAHA (2025–26): Science and Humanities Association - organized academic events.
- Member, ARME (2024 – Present): Association of Robotics and Mechatronics Engineers.
- YRC Volunteer (2024 – Present): Organized Youth Fest events for visually challenged individuals.
- Assistant School Pupil Leader (2019 – 20): Exercised leadership and administrative support for student body operations and school-wide event coordination.
- Master of Ceremonies (MC) (2017 – 2020): Served as a professional host for multiple school events, developing public speaking and audience engagement skills.

## LANGUAGES

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English (Proficient) | Tamil (Proficient) | Telugu (Conversational) | Japanese (Elementary) | Hindi (Elementary)